A

PROJECT REPORT

On

FailSight: Predictive Maintenance and Reliability Intelligence System for Turbofan Engines Using Machine Learning

Submitted In Partial Fulfillment of the Requirements
for the Degree of
Bachelor of Technology
Computer Science and Engineering
of
FACULTY OF ENGINEERING AND TECHNOLOGY
by

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BONAFIDE CERTIFICATE

Certified that this project report titled **"FailSight: Predictive Maintenance and Reliability Intelligence System for Turbofan Engines Using Machine Learning "** is the Bonafide work of **PRAJJAVAL PANDEY [RA2311003030542], DIVYA TRIPATHI [RA2311003030547], JIYA CHAUDHARY [RA2311003030560]**, who carried out the project work under my supervision. Certified further, that to the best of my knowledge, the work reported herein does not form any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE	**SIGNATURE**
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DECLARATION

We, **Prajjaval Pandey (RA2311003030542), Divya Tripathi (RA2311003030547), and Jiya Chaudhary (RA2311003030560)** hereby declare that the work which is being presented in the project report titled "**FailSight: Predictive Maintenance and Reliability Intelligence System for Turbofan Engines Using Machine Learning**" is the record of authentic work carried out by us during the period from January to May,2026 and submitted by us in partial fulfillment for the award of the degree "Bachelor of Technology in Computer Science and Engineering" to SRM Institute of Science and Technology, Delhi-NCR Campus, Modinagar, U.P.(INDIA).This work has not been submitted to any other University or Institute for the award of any Degree/Diploma.

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ABSTRACT

With the growing demand for reliable industrial systems, predictive maintenance has become essential for reducing unexpected failures and improving operational efficiency. This project presents **FailSight**, an end-to-end predictive maintenance and reliability intelligence system designed to predict equipment health and Remaining Useful Life (RUL) using machine learning techniques.

The system utilizes the NASA CMAPSS turbofan engine dataset to analyze sensor data and identify degradation patterns. A robust data pipeline performs data preprocessing and feature engineering, including rolling statistics, lag features, and anomaly scores. A LightGBM model is implemented for RUL prediction, while an Isolation Forest algorithm is used for anomaly detection. Additionally, Weibull-based reliability analysis is conducted to estimate failure probability and Mean Time to Failure (MTTF).

An interactive dashboard built using Streamlit provides real-time visualization of fleet health, anomaly trends, and reliability insights. The system also integrates an AI-based assistant for natural language queries and automated reporting. Overall, **FailSight** improves maintenance planning, reduces downtime, and enhances system reliability through data-driven decision-making.

CHAPTER 1
INTRODUCTION

1.1 Background

In modern industries such as aviation, manufacturing, and energy systems, maintaining equipment reliability is critical for ensuring safety, productivity, and cost efficiency. Unexpected machine failures can lead to downtime, increased operational costs, and safety risks. Traditional maintenance strategies such as reactive maintenance (repair after failure) and preventive maintenance (scheduled servicing) are widely used but often inefficient. Reactive maintenance causes sudden disruptions, while preventive maintenance may lead to unnecessary servicing even when equipment is functioning properly.

With the advancement of sensor technologies and data analytics, predictive maintenance has emerged as an effective approach to monitor equipment health and predict failures before they occur. By analyzing historical and real-time sensor data, predictive maintenance systems estimate the Remaining Useful Life (RUL) of machines and enable timely maintenance decisions.

1.2 Existing Solutions

Existing predictive maintenance systems commonly use threshold-based monitoring and statistical methods to detect abnormal conditions. These systems generate alerts when sensor values exceed predefined limits. While simple to implement, such approaches are limited in their ability to analyze complex relationships between multiple sensor readings.

More advanced solutions incorporate machine learning techniques to predict equipment degradation and detect anomalies. Some systems also use reliability analysis methods such as Weibull distribution to estimate failure probability. However, many existing solutions focus only on prediction or anomaly detection individually, rather than providing a fully integrated platform that combines data processing, prediction, reliability analysis, and visualization.

1.3 Challenges in Existing Systems

Despite advancements in predictive maintenance technologies, several limitations still exist in current systems. These challenges affect prediction accuracy, scalability, and usability in real-world industrial environments.

The major challenges include:

- **Limited Early Fault Detection:**
  Many systems fail to detect early-stage degradation because minor changes in sensor readings are difficult to distinguish from normal variations.
- **Insufficient Feature Engineering:**
  Basic models often rely on limited features and do not capture temporal patterns such as trends, cycles, and degradation behavior over time.
- **Lack of Integrated Reliability Metrics:**
  Existing systems often predict failures but do not provide reliability measures such as Mean Time to Failure (MTTF) or hazard rate analysis.

- **Handling Large-Scale Sensor Data:**
  Industrial datasets contain large volumes of sensor readings, making efficient data processing and storage a major challenge.
- **Poor Visualization and Decision Support:**
  Many systems lack interactive dashboards, making it difficult for users to interpret results and take timely maintenance actions.
- **Manual Maintenance Planning:**
  Without automated insights and recommendations, maintenance teams rely on manual judgment, increasing the risk of errors and delays.

1.4 Proposed Solution: FailSight

To address the challenges identified in existing predictive maintenance systems, this project introduces **FailSight**, an end-to-end predictive maintenance and reliability intelligence platform. The proposed system integrates machine learning, anomaly detection, reliability analysis, and visualization into a unified framework.

The key components of the proposed solution include:

- **Automated Data Pipeline:**
  Implements data ingestion, cleaning, and preprocessing to ensure consistent and reliable input data for analysis.
- **Advanced Feature Engineering:**
  Generates meaningful features such as rolling statistics, temporal lag features, cycle ratios, and degradation indices to improve prediction accuracy.
- **Machine Learning-Based RUL Prediction:**
  Uses a LightGBM model to accurately estimate Remaining Useful Life (RUL) of equipment based on historical sensor data.
- **Anomaly Detection System:**
  Applies Isolation Forest algorithms to identify abnormal sensor behavior and detect early warning signals of equipment failure.
- **Reliability Analysis Module:**
  Utilizes Weibull distribution models to estimate failure probability, hazard rates, and Mean Time to Failure (MTTF).
- **Interactive Visualization Dashboard:**
  Provides real-time insights through graphs, heatmaps, and performance indicators using Streamlit and Plotly.
- **AI-Powered Insight Generation:**
  Integrates a Retrieval-Augmented Generation (RAG) assistant to enable natural language queries and automated maintenance reporting.
- **Automated Scheduling and Monitoring:**
  Uses scheduled pipelines to perform daily updates, ensuring continuous monitoring and system reliability.

Overall, the proposed **FailSight** system offers a scalable, integrated, and intelligent solution that improves maintenance planning, reduces unexpected failures, and enhances operational efficiency.

CHAPTER 2
LITERATURE SURVEY

Predictive maintenance has become an important research area due to the increasing need for reliable industrial systems and reduced operational downtime. Several researchers have explored statistical, machine learning, and deep learning techniques to estimate equipment health and Remaining Useful Life (RUL). A comprehensive review by **Lei et al. [1]** presented a systematic overview of machinery health prognostics, covering data acquisition, feature engineering, and RUL prediction methods. Their study highlighted the importance of data-driven approaches in improving prediction accuracy compared to traditional maintenance strategies.

Statistical reliability methods, particularly Weibull distribution models, have been widely used to estimate failure probability and equipment lifespan. These methods provide valuable reliability metrics such as Mean Time to Failure (MTTF) and hazard rates. However, traditional statistical models often struggle to capture nonlinear relationships in large multi-sensor datasets. To address these limitations, researchers introduced machine learning-based approaches for predictive maintenance. **Wang et al. [2]** proposed a functional data analysis method for RUL estimation, demonstrating improved performance on complex datasets such as NASA CMAPSS turbofan engine data.

Further advancements in predictive maintenance were achieved through the use of ensemble learning and deep learning techniques. **Zhu et al. [3]** introduced a multi-task learning framework that simultaneously predicts multiple degradation patterns, resulting in improved accuracy and robustness. Similarly, **Wang et al. [4]** developed a deep separable convolutional network that effectively captures temporal dependencies in sensor data for accurate RUL prediction. These deep learning models have shown promising results but often require large computational resources and extensive training data.

Anomaly detection techniques have also played a significant role in predictive maintenance research. Early fault detection methods using unsupervised learning models help identify abnormal patterns before system failure occurs. **Sánchez et al. [5]** demonstrated the effectiveness of LSTM-based autoencoders in detecting early-stage faults in turbofan engine datasets. Additionally, Isolation Forest and other unsupervised methods have been widely used due to their scalability and efficiency in high-dimensional sensor data environments.

The proposed **FailSight** system builds upon previous research findings by integrating machine learning-based RUL prediction, anomaly detection using Isolation Forest, Weibull reliability analysis, and interactive visualization into a single platform. By combining automated data pipelines with advanced analytics and visualization tools, the system aims to improve prediction accuracy, enhance equipment reliability, and support efficient maintenance decision-making.

CHAPTER 3

Existing Problem and Proposed Solution

Existing Problem

In many industrial environments, equipment maintenance is still based on traditional methods such as reactive or preventive maintenance. These approaches often fail to accurately predict failures, leading to unexpected downtime and increased operational costs. Although predictive maintenance solutions exist, many of them lack integration, scalability, and real-time decision support capabilities.

The major problems in existing systems include:

- **Reactive Maintenance Dependency:**
  Many industries still rely on repairing equipment only after failure occurs, which leads to sudden downtime and high repair costs.
- **Inefficient Preventive Maintenance:**
  Scheduled maintenance is performed at fixed intervals, even when equipment is functioning normally, resulting in unnecessary maintenance expenses.
- **Limited Prediction Accuracy:**
  Existing models often rely on basic statistical methods or limited features, reducing the accuracy of Remaining Useful Life (RUL) predictions.
- **Lack of Early Anomaly Detection:**
  Traditional monitoring systems struggle to detect early-stage faults due to minor variations in sensor data.
- **Absence of Reliability Insights:**
  Many systems do not provide reliability metrics such as failure probability, hazard rate, or Mean Time to Failure (MTTF).
- **Poor Visualization and Monitoring:**
  Existing tools often lack interactive dashboards for real-time visualization of machine health and performance trends.
- **Manual Data Processing:**
  In many systems, data handling and model execution require manual intervention, making the process time-consuming and prone to errors.

Proposed Solution

To overcome the limitations of existing systems, this project proposes **FailSight**, an integrated predictive maintenance and reliability intelligence platform designed to improve equipment monitoring and failure prediction accuracy. The system combines machine learning, anomaly detection, reliability analysis, and visualization into a unified framework.

The proposed solution includes the following key features:

- **Automated Data Pipeline:**
  Implements automated data ingestion, preprocessing, and storage to efficiently handle large-scale sensor datasets.
- **Advanced Feature Engineering:**
  Generates meaningful features such as rolling statistics, lag features, cycle ratios, and degradation indices to capture machine behavior patterns.
- **Accurate RUL Prediction:**
  Uses a LightGBM machine learning model to predict Remaining Useful Life (RUL) with improved accuracy.
- **Early Anomaly Detection:**
  Applies Isolation Forest algorithms to identify abnormal sensor readings and detect potential failures at an early stage.
- **Reliability Analysis Integration:**
  Utilizes Weibull distribution models to estimate failure probability, hazard rate, and Mean Time to Failure (MTTF).
- **Interactive Visualization Dashboard:**
  Provides real-time insights through graphs, KPIs, heatmaps, and alerts using Streamlit and Plotly.
- **AI-Based Query System:**
  Integrates a Retrieval-Augmented Generation (RAG) assistant to allow users to query system health and generate automated reports.
- **Automated Scheduling:**
  Uses scheduled workflows to run the pipeline periodically, ensuring continuous monitoring and updated predictions.

Overall, the proposed **FailSight** system provides an intelligent, scalable, and user-friendly predictive maintenance solution that reduces downtime, improves maintenance efficiency, and enhances equipment reliability.

Functional Requirements

Functional requirements describe the specific operations and tasks that the system must perform to achieve its objectives. The **FailSight** system includes the following functional requirements:

- **Data Ingestion:**
  The system shall collect and import turbofan engine sensor data from the NASA CMAPSS dataset into the processing environment.
- **Data Preprocessing:**
  The system shall clean raw data by handling missing values, formatting data, and organizing it into structured datasets suitable for analysis.
- **Feature Engineering:**
  The system shall generate additional features such as rolling mean, rolling standard deviation, lag features, cycle ratio, and degradation index.
- **Remaining Useful Life (RUL) Prediction:**
  The system shall predict the Remaining Useful Life of turbofan engines using a trained LightGBM machine learning model.
- **Anomaly Detection:**
  The system shall detect abnormal sensor readings using Isolation Forest algorithms.
- **Reliability Analysis:**
  The system shall estimate reliability metrics such as failure probability, hazard rate, and Mean Time to Failure (MTTF) using Weibull distribution.
- **Spare Parts Forecasting:**
  The system shall predict maintenance needs and spare part requirements based on predicted failure conditions.
- **Dashboard Visualization:**
  The system shall provide graphical visualizations such as histograms, line charts, heatmaps, and performance indicators using an interactive dashboard.
- **AI-Based Query System:**
  The system shall allow users to interact with the system using natural language queries through an AI-powered assistant.
- **Automated Scheduling:**
  The system shall execute predefined workflows automatically at scheduled intervals.

Non-Functional Requirements

Non-functional requirements describe the performance, usability, reliability, and quality attributes of the system.

- **Performance:**
  The system should process large volumes of sensor data efficiently and generate predictions within a reasonable time.
- **Scalability:**
  The system should support increasing data volumes and additional sensor inputs without performance degradation.
- **Reliability:**
  The system should provide consistent and accurate predictions with minimal failure or downtime.
- **Usability:**
  The system should provide a user-friendly interface with clear visualization tools and easy navigation.
- **Maintainability:**
  The system should allow easy updates, debugging, and modification of components such as models and pipelines.
- **Security:**
  The system should ensure safe storage and processing of data to prevent unauthorized access.
- **Availability:**
  The system should remain accessible and functional for users whenever required.
- **Accuracy:**
  The system should provide precise predictions with minimal error rates.

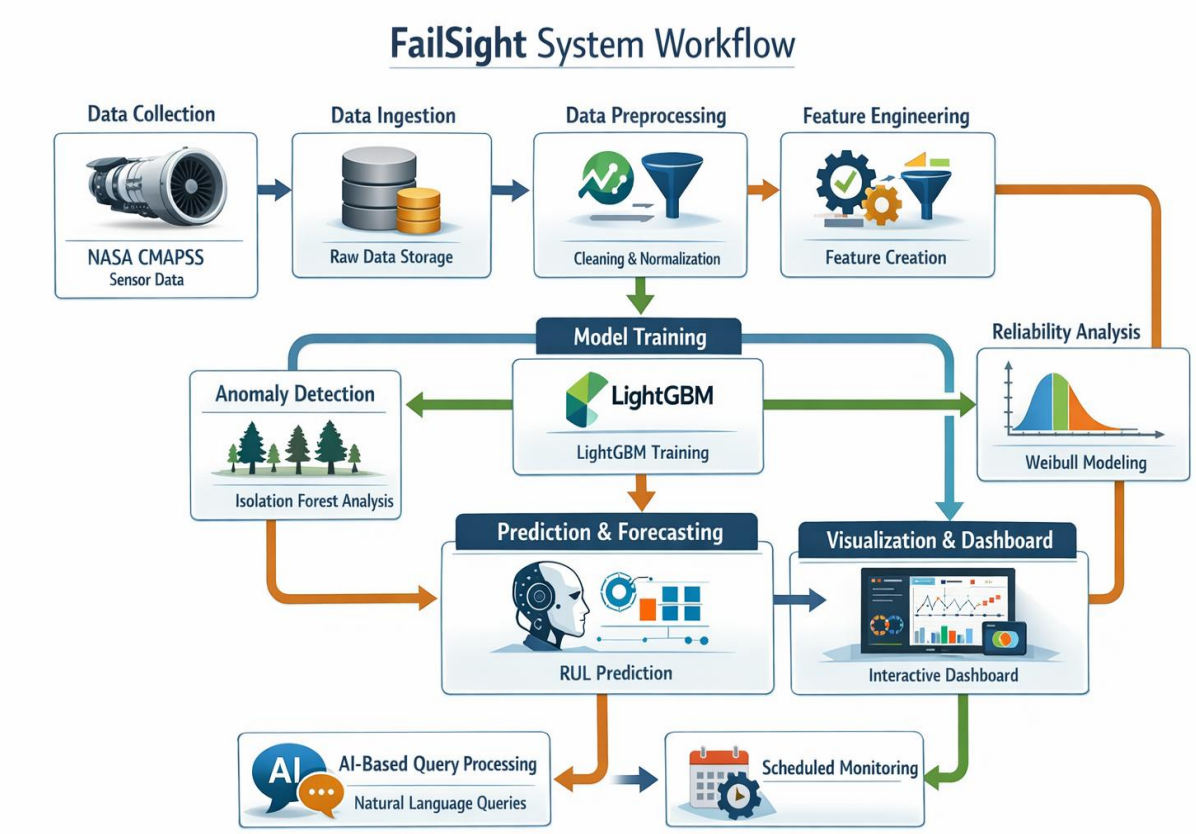
System Workflow

The **FailSight** system follows a structured workflow that processes sensor data through multiple stages to generate predictions and insights. The workflow ensures efficient handling of data, model training, prediction, and visualization.

The system workflow includes the following steps:

1. **Data Collection**
   The system collects turbofan engine sensor data from the NASA CMAPSS dataset.
2. **Data Ingestion**
   Raw data is imported into the system and stored in a structured format such as a database or processed files.
3. **Data Preprocessing**
   The data is cleaned by removing inconsistencies, handling missing values, and normalizing sensor readings.
4. **Feature Engineering**
   New features such as rolling statistics, lag values, and degradation metrics are generated to improve model performance.
5. **Model Training**
   The LightGBM machine learning model is trained using historical sensor data and labeled Remaining Useful Life values.

6. **Prediction and Forecasting**
   The trained model predicts Remaining Useful Life and forecasts potential maintenance requirements.
7. **Visualization and Dashboard Display**
   Results are displayed through an interactive dashboard showing system health, anomalies, and reliability insights.
8. **AI-Based Query Processing**
   Users can interact with the system using natural language queries to generate automated reports and insights.
9. **Automated Scheduling**
   The entire workflow runs automatically at scheduled intervals to ensure continuous monitoring and updates.



CHAPTER 4
SYSTEM DESIGN

Overview of System Design

The system design of **FailSight** focuses on creating a scalable and modular predictive maintenance platform capable of handling large-scale sensor data and generating accurate maintenance insights. The system is designed using a pipeline-based architecture where each component performs a specific task such as data ingestion, preprocessing, feature engineering, model training, anomaly detection, reliability analysis, and visualization. This modular design ensures flexibility, maintainability, and efficient data processing.

The system integrates machine learning models, statistical analysis techniques, and visualization tools to provide an end-to-end predictive maintenance solution. The overall design ensures smooth data flow between components and enables automated execution of the maintenance workflow.

System Architecture Design

The **FailSight** system follows a layered architecture consisting of multiple interconnected modules. Each module performs a specific function in the predictive maintenance pipeline.

The main components of the system architecture include:

1. Data Source Layer

This layer provides the input data required for system operation.

- NASA CMAPSS turbofan engine dataset
- Multi-sensor telemetry data
- Historical operational cycle data

This layer acts as the foundation of the system by supplying raw sensor readings for further processing.

2. Data Processing Layer

This layer is responsible for transforming raw data into structured and meaningful datasets.

- **Data Ingestion Module:**
  Loads raw sensor data and stores it in structured formats.
- **Data Preprocessing Module:**
  Cleans the data by handling missing values, removing noise, and normalizing sensor readings.
- **Feature Engineering Module:**
  Generates additional features such as rolling statistics, lag features, cycle ratio, and degradation index.

This layer ensures that the dataset is suitable for machine learning analysis.

3. Machine Learning Layer

This layer performs prediction and anomaly detection tasks.

- **RUL Prediction Module:**
  Uses a LightGBM model to estimate Remaining Useful Life (RUL) of turbofan engines.
- **Anomaly Detection Module:**
  Uses Isolation Forest algorithms to identify abnormal sensor behavior.
- **Model Training and Evaluation Module:**
  Trains machine learning models using historical data and evaluates performance using metrics such as RMSE and R^2.

This layer forms the core intelligence of the system.

4. Reliability Analysis Layer

This layer performs statistical reliability analysis to estimate equipment lifespan.

- **Weibull Analysis Module:**
  Calculates reliability metrics such as failure probability, hazard rate, and Mean Time to Failure (MTTF).
- **Spare Parts Forecasting Module:**
  Predicts maintenance requirements based on predicted failures.

This layer helps in long-term maintenance planning.

5. Visualization and User Interface Layer

This layer provides graphical outputs and system interaction capabilities.

- **Dashboard Module:**
  Displays visual insights such as charts, graphs, histograms, and heatmaps using Streamlit and Plotly.
- **AI Query Interface:**
  Allows users to interact with the system using natural language queries and generate automated reports.

This layer enhances usability and decision-making.

6. Automation Layer

This layer ensures continuous system execution.

- **Scheduler Module:**
  Uses automated scheduling tools to execute data pipelines at predefined intervals.
- **Logging and Monitoring Module:**
  Records system activities and errors for debugging and performance tracking.

This layer ensures reliability and operational efficiency.

Data Flow Design

The data flow within the FailSight system follows a structured sequence:

1. Raw sensor data is collected from the dataset.
2. Data is ingested and stored in structured format.
3. Preprocessing is applied to clean and normalize the data.
4. Feature engineering generates additional meaningful variables.
5. Machine learning models predict Remaining Useful Life.
6. Anomaly detection identifies abnormal patterns.
7. Reliability analysis calculates failure probabilities.
8. Results are displayed on the interactive dashboard.
9. Automated scheduling ensures periodic system execution.

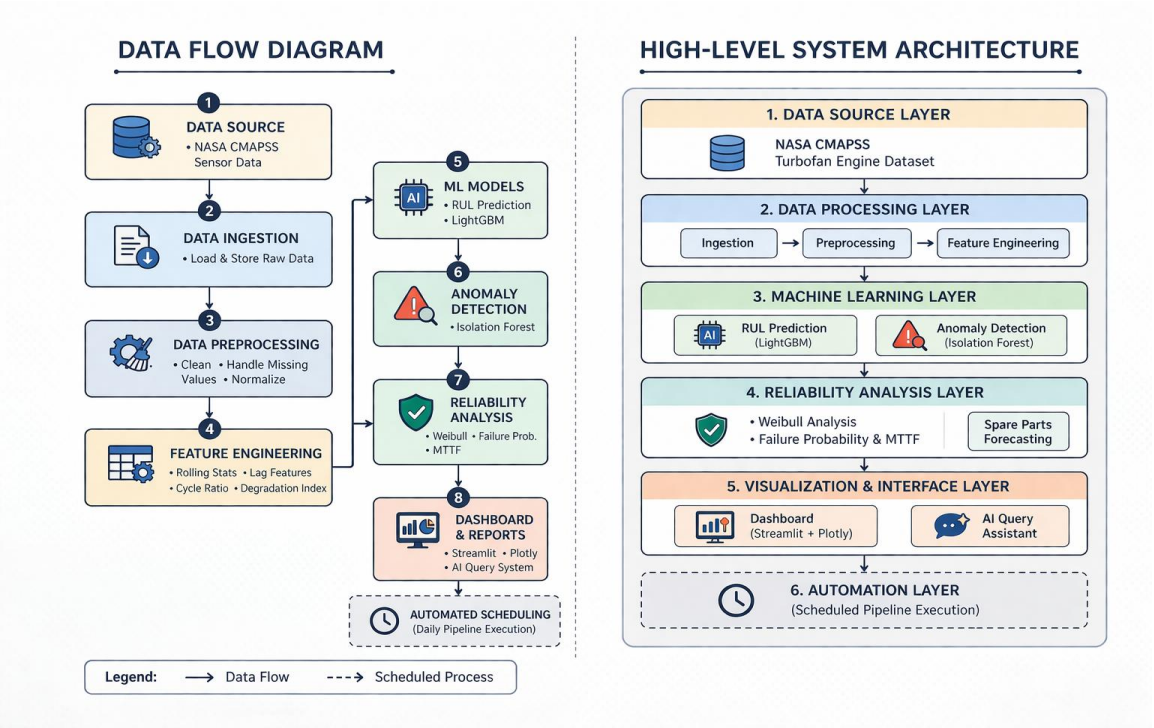
This sequential data flow ensures accurate processing and timely generation of maintenance insights.

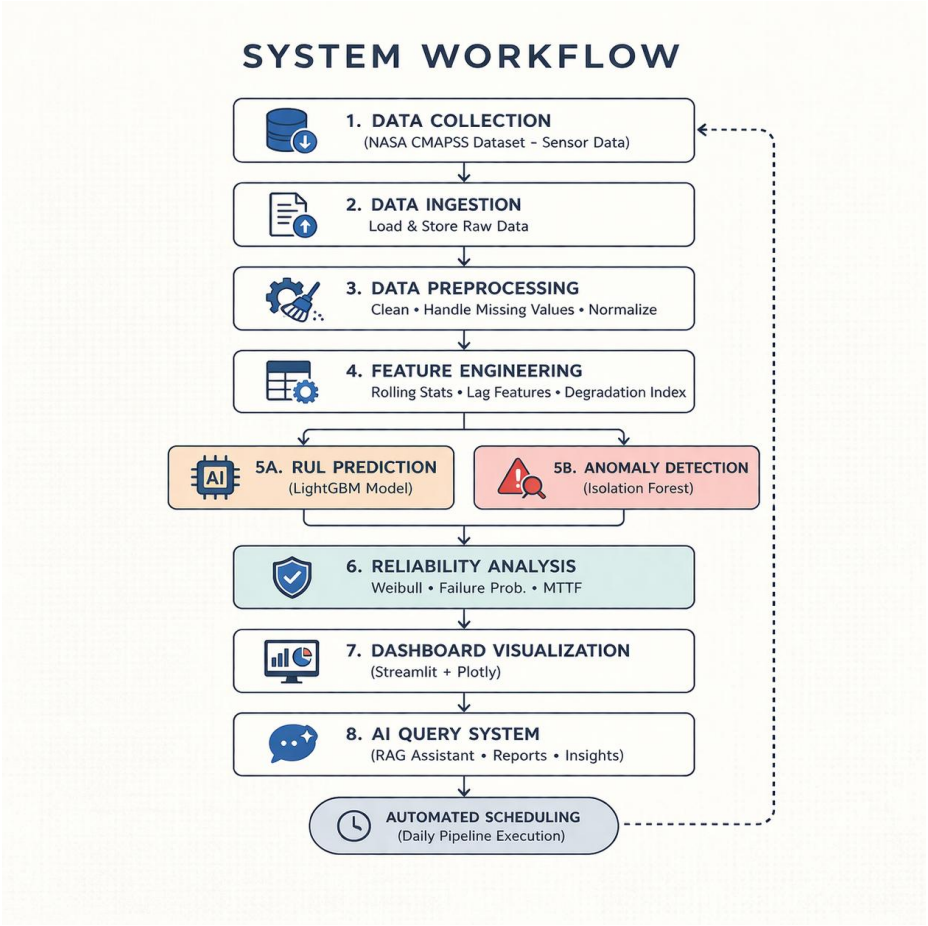
Module Design

The FailSight system is divided into several functional modules:

- **Ingestion Module** — Handles data loading and storage
- **Feature Engineering Module** — Creates derived features
- **Model Training Module** — Builds prediction models
- **Anomaly Detection Module** — Identifies abnormal behavior
- **Reliability Module** — Performs statistical reliability analysis
- **Visualization Module** — Displays graphical insights
- **Automation Module** — Executes scheduled workflows

This modular design improves maintainability and simplifies future enhancements


DATA FLOW DIAGRAM
1
DATA SOURCE
• NASA CMAPSS Sensor Data
2
DATA INGESTION
• Load & Store Raw Data
3
DATA PREPROCESSING
• Clean • Handle Missing Values • Normalize
4
FEATURE ENGINEERING
• Rolling Stats • Lag Features
• Cycle Ratio • Degradation Index
5
ML MODELS
• RUL Prediction
• LightGBM
6
ANOMALY DETECTION
• Isolation Forest
7
RELIABILITY ANALYSIS
• Weibull • Failure Prob.
• MTTF
8
DASHBOARD & REPORTS
• Streamlit • Plotly
• AI Query System
AUTOMATED SCHEDULING
(Daily Pipeline Execution)
Legend: Data Flow Scheduled Process
HIGH-LEVEL SYSTEM ARCHITECTURE
1. DATA SOURCE LAYER
NASA CMAPSS
Turbofan Engine Dataset
2. DATA PROCESSING LAYER
Ingestion
Preprocessing
Feature Engineering
3. MACHINE LEARNING LAYER
RUL Prediction
(LightGBM)
Anomaly Detection
(Isolation Forest)
4. RELIABILITY ANALYSIS LAYER
• Weibull Analysis
• Failure Probability & MTTF
Spare Parts Forecasting
5. VISUALIZATION & INTERFACE LAYER
Dashboard
(Streamlit + Plotly)
AI Query Assistant
6. AUTOMATION LAYER
(Scheduled Pipeline Execution)


SYSTEM WORKFLOW
1. DATA COLLECTION
(NASA CMAPSS Dataset - Sensor Data)
2. DATA INGESTION
Load & Store Raw Data
3. DATA PREPROCESSING
Clean • Handle Missing Values • Normalize
4. FEATURE ENGINEERING
Rolling Stats • Lag Features • Degradation Index
5A. RUL PREDICTION
(LightGBM Model)
5B. ANOMALY DETECTION
(Isolation Forest)
6. RELIABILITY ANALYSIS
Weibull • Failure Prob. • MTTF
7. DASHBOARD VISUALIZATION
(Streamlit + Plotly)
8. AI QUERY SYSTEM
(RAG Assistant • Reports • Insights)
AUTOMATED SCHEDULING
(Daily Pipeline Execution)

CHAPTER 5
CODE AND OUTPUT

The FailSight system was implemented as a modular end-to-end predictive maintenance pipeline that integrates data engineering, machine learning, and reliability analysis. The system is designed with a component-based architecture to ensure scalability, maintainability, and clear separation of concerns across the workflow.

Code Implementation Overview

1. Data Pipeline (pipelines/)

The data pipeline is responsible for ingesting and preprocessing the NASA CMAPSS turbofan engine dataset.

- Loads raw multivariate sensor data for multiple engine units
- Performs data cleaning, normalization, and preprocessing
- Generates Remaining Useful Life (RUL) labels for supervised learning
- Stores processed data in structured formats such as SQLite and Parquet

This stage ensures that raw operational sensor data is transformed into a machine-learning-ready format with consistent temporal alignment across engine cycles.

2. Feature Engineering (pipelines/features.py)

This module transforms raw sensor readings into meaningful predictive features that capture degradation behavior.

- Rolling statistical features such as mean and standard deviation over defined windows
- Temporal lag features to capture sequential dependencies in sensor behavior
- Engine health indicators including degradation index and cycle ratio
- Anomaly detection scores using Isolation Forest

These engineered features enhance model performance by representing both short-term fluctuations and long-term degradation trends in engine health.

3. Model Training (models/train_rul.py)

A LightGBM regression model is trained to predict Remaining Useful Life (RUL).

- Dataset is divided into training, validation, and testing splits
- The model learns nonlinear relationships between sensor patterns and engine degradation
- Evaluation is performed using RMSE, MAE, and R^2 score
- The trained model is saved using joblib for reuse in inference and deployment

This model serves as the core predictive component of the system.

4. Reliability Analysis (analysis/weibull_analysis.py)

This module applies statistical reliability engineering methods to model engine failure behavior.

- Fits Weibull distribution to observed failure time data
- Estimates shape parameter (β), which indicates failure behavior (early-life, random, or wear-out)
- Estimates scale parameter (α), representing characteristic life of the system
- Computes Mean Time To Failure (MTTF)
- Generates hazard rate curves to analyze how failure probability evolves over time

This provides a theoretical reliability perspective that complements machine learning predictions.

Output Generated

The system generates two primary outputs: an interactive dashboard and a reliability analysis report.

1. Interactive Dashboard (Streamlit)

A multi-page dashboard is developed to provide real-time insights into fleet health and predictive analytics.

Key Components:

Fleet Overview

- Total number of engine units monitored
- Distribution of engines across healthy, warning, and critical categories
- Average Remaining Useful Life across the fleet
- Alert table highlighting high-risk engines

RUL Analytics

- Distribution of predicted RUL values
- Threshold-based segmentation of engine health states
- Visualization of degradation trends across engine lifecycles

Exploratory Data Analysis

- Sensor-wise distribution plots
- Correlation analysis between features and RUL
- Time-series visualization of sensor degradation patterns

Unit-Level Analysis

- Individual engine monitoring with cycle-wise tracking
- Sensor trend visualization for selected units
- Degradation index progression over time

Anomaly Analysis

- Identification of anomalous engine behavior using Isolation Forest
- Heatmaps showing anomaly distribution across engines
- Relationship analysis between anomalies and RUL degradation

The dashboard enables both high-level fleet monitoring and detailed diagnostic investigation at the unit level.

2. Weibull Reliability Analysis Report

A statistical reliability report is generated to analyze failure behavior across the engine fleet using Weibull distribution modeling.

Key Outputs:

Weibull Parameter Estimation

Shape parameter (β) indicating failure mode characteristics

Scale parameter (α) representing characteristic lifetime of the engine population

Mean Time To Failure (MTTF)

Provides an estimate of the average operational lifespan of engines

Hazard Rate Analysis

Models failure probability as a function of time

Identifies periods of increasing or accelerating failure risk

Subset-Level Comparison (FD001–FD004)

Reliability comparison across different operational conditions

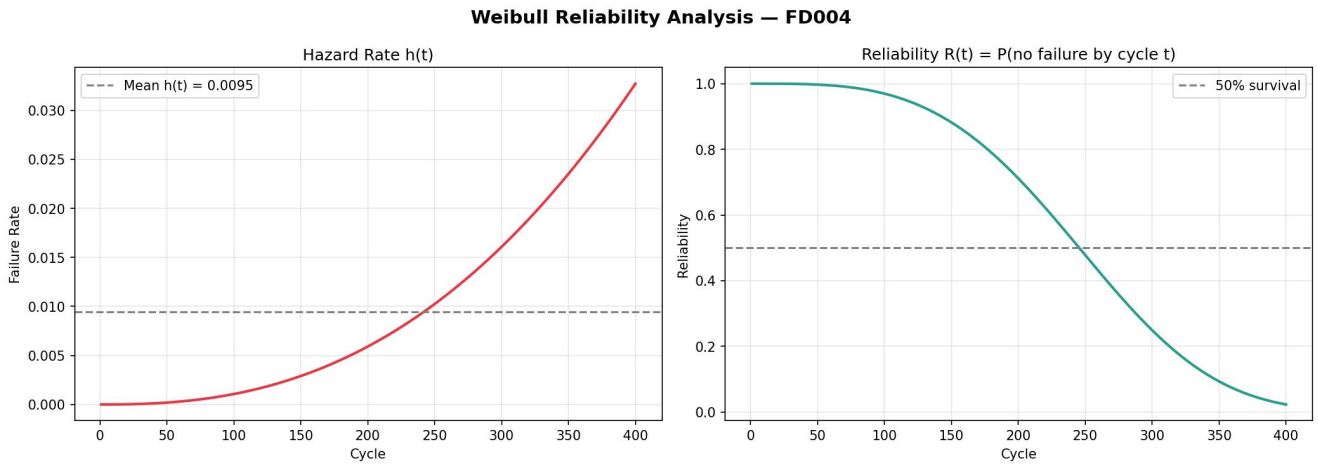
Analysis of how environmental stress impacts degradation patterns

Final Output Summary

Component	Description
Dashboard	Interactive monitoring of fleet health, RUL prediction, anomaly detection, and exploratory analysis
Weibull Report	Statistical reliability modeling and interpretation of failure behavior

Conclusion

The system integrates machine learning-based predictive maintenance with classical reliability engineering techniques. The machine learning component provides data-driven predictions of Remaining Useful Life, while the Weibull analysis offers a statistical interpretation of system failure behavior.



CHAPTER 6

CONCLUSION

The FailSight system demonstrates an end-to-end predictive maintenance framework that combines machine learning, feature engineering, and reliability analysis to assess and predict turbofan engine health using the NASA CMAPSS dataset.

The machine learning component, built using a LightGBM regression model, effectively predicts Remaining Useful Life (RUL) by learning complex nonlinear relationships from multivariate sensor data. This enables early identification of degradation patterns and supports proactive maintenance decisions.

In parallel, the system incorporates classical reliability engineering through Weibull analysis, providing statistical insights into engine failure behavior. Parameters such as shape and scale, along with derived metrics like Mean Time To Failure (MTTF) and hazard rates, help in understanding long-term reliability trends across different operating conditions.

The interactive Streamlit dashboard further enhances usability by enabling real-time visualization of fleet health, anomaly detection, and unit-level diagnostics. It allows users to explore both high-level fleet summaries and detailed sensor-level behavior.

Overall, the integration of predictive modeling with reliability analysis results in a comprehensive decision-support system. It not only forecasts failures but also explains underlying degradation behavior, making the system valuable for industrial predictive maintenance applications.

CHAPTER 7

FUTURE ENHANCEMENTS

While the current FailSight system provides a comprehensive predictive maintenance and reliability analysis framework, several enhancements can further improve its accuracy, scalability, and real-world applicability.

1. Advanced Deep Learning Models

- Integrate sequence-based models such as LSTM, GRU, and Transformer architectures for improved RUL prediction.
- Explore attention-based models to better capture long-term temporal dependencies in sensor data.
- Compare performance against LightGBM to evaluate trade-offs between accuracy and interpretability.

2. Multi-Step Ahead Failure Forecasting

- Extend the current single-point RUL prediction to multi-step forecasting of engine degradation.
- Enable prediction of failure trajectories over future cycles rather than only final RUL values.

3. Enhanced Anomaly Detection

- Incorporate hybrid anomaly detection techniques combining statistical methods and deep learning (e.g., Autoencoders).
- Improve early-warning detection by reducing false positives in high-noise sensor environments.

4. Real-Time Streaming Pipeline

- Replace batch processing with real-time data ingestion using streaming frameworks such as Kafka.
- Enable continuous monitoring of engine sensor data for live predictive maintenance.

5. Model Interpretability and Explainability

- Integrate SHAP or LIME to explain feature contributions to RUL predictions.
- Provide explainable insights in the dashboard for better trust and decision-making in industrial settings.

6. Cloud Deployment and Scalability

- Deploy the system on cloud platforms such as AWS, Azure, or GCP.
- Use containerization (Docker) and orchestration (Kubernetes) for scalable deployment.
- Enable API-based inference for integration with external maintenance systems.

7. Automated Alerting System

- Implement notification systems using email, Slack, or SMS for critical engine conditions.
- Trigger alerts based on RUL thresholds or anomaly detection outputs.

8. Continuous Learning Pipeline

- Introduce incremental learning to update models as new sensor data becomes available.
- Reduce model drift and improve long-term prediction stability.

CHAPTER 8

REFERENCES

1] A. Saxena, K. Goebel, D. Simon, and N. Eklund, "Damage Propagation Modeling for Aircraft Engine Run-to-Failure Simulation," NASA Ames Prognostics Data Repository, 2008. Available: https://ti.arc.nasa.gov/tech/dash/groups/pcoe/prognostic-data-repository/

[2] Y. Lei, N. Li, L. Guo, N. Li, T. Yan, and J. Lin, "Machinery Health Prognostics: A Systematic Review of Machine Learning Methods," *IEEE Transactions on Reliability*, vol. 65, no. 3, pp. 1–18, 2016.

[3] G. Ke, Q. Meng, T. Finley, et al., "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

[4] F. T. Liu, K. M. Ting, and Z. H. Zhou, "Isolation Forest," in *Proceedings of the IEEE International Conference on Data Mining (ICDM)*, 2008, pp. 413–422.

[5] W. Weibull, "A Statistical Distribution Function of Wide Applicability," *Journal of Applied Mechanics*, vol. 18, no. 3, pp. 293–297, 1951.

[6] Streamlit Inc., "Streamlit Documentation," Available: https://docs.streamlit.io

[7] F. Pedregosa et al., "Scikit-learn: Machine Learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[8] Plotly Technologies Inc., "Plotly Python Documentation," Available: https://plotly.com/python/

[9] LangChain AI, "LangChain Documentation," Available: https://docs.langchain.com

[10] LangChain AI, "LangGraph Documentation," Available: https://langchain-ai.github.io/langgraph/

[11] Chroma AI, "ChromaDB Documentation," Available: https://docs.trychroma.com/

[12] W. McKinney, "Data Structures for Statistical Computing in Python," in *Proceedings of the 9th Python in Science Conference*, 2010.